

Theme 2: Limits, Continuity & Derivatives

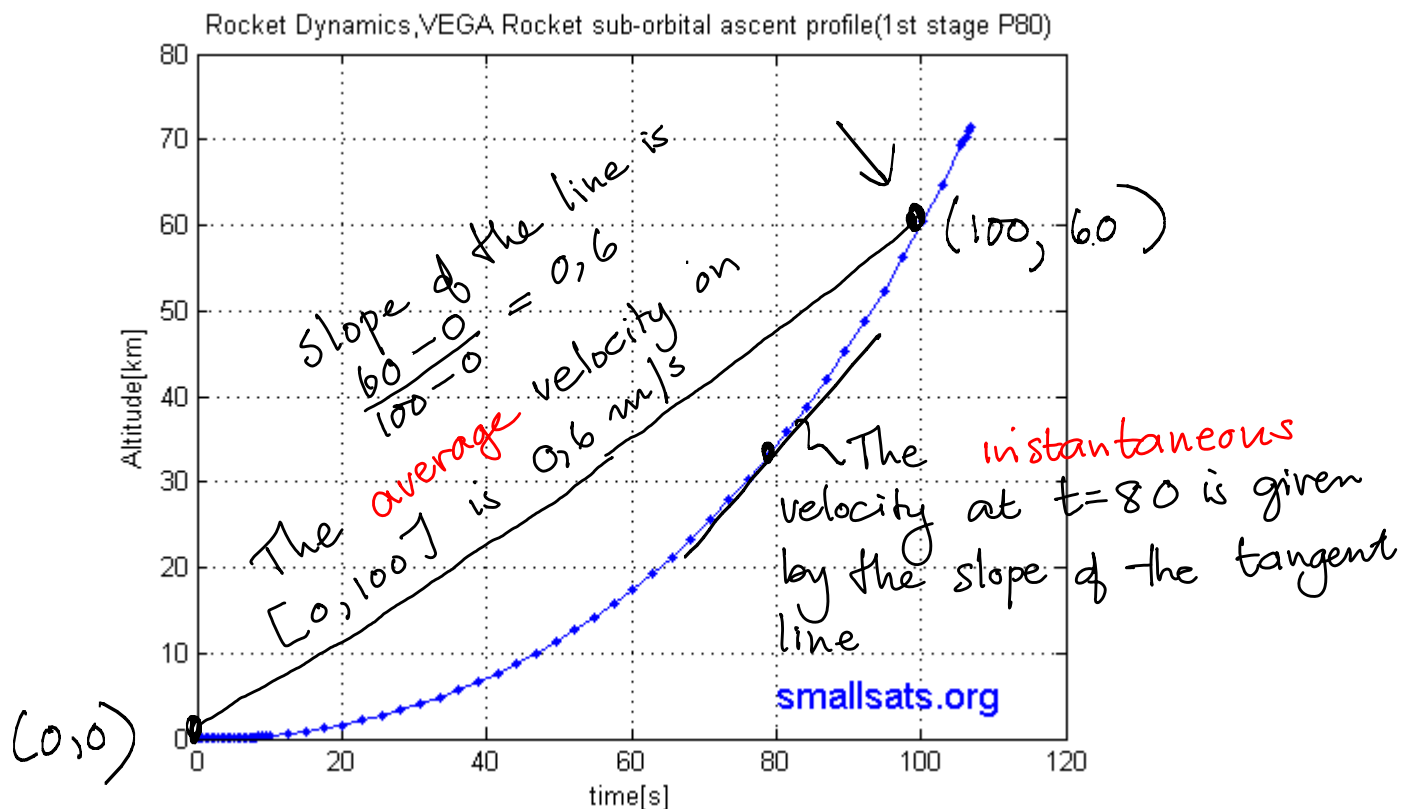
Unit 2.1 Tangent & velocity problems pp 81-86.

Note Title

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Difference between average velocity and instantaneous velocity.

Example:



Ave velocity over the first 100 sec: $\frac{60-0}{100-0} = 0,6 \text{ km/s}$

To estimate the instantaneous velocity at $t=80$

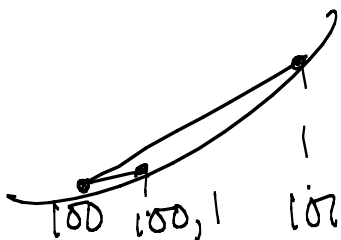
Say $f(t) = 0,006 t^2$.

Consider the interval $[80, 81]$

$$\frac{0,006(81)^2 - 0,006(80)^2}{81 - 80} = 0,966 \text{ m/s}$$

Better estimate: Consider the interval $[80, 80.1]$

$$\frac{0,006(80.1)^2 - 0,006(80)^2}{80.1 - 80} = 0,9606 \text{ m/s}$$



By taking smaller and smaller intervals around 80, we get a better approximation for the instantaneous velocity at $t=80$.

Through the process of differentiation we will see that the instantaneous velocity is given by $f'(t) = 0,012t$. At $t=80$ the instantaneous velocity is $0,96 \text{ m/s}$.

Unit 2.1 is introductory and illustrates the difference between average velocity over an interval and instantaneous velocity at a point.

In general velocity is an example of rate of change, so we speak of average rate of change and instantaneous rate of change.